Chapter B, Problem 4PP

Problem

Evaluate these complex fractions:

(a) $\dfrac{6\underline{/30^\circ} + j5 - 3}{-1 + j + 2e^{j45^\circ}}$ (b) $\left[\dfrac{(15 - j7)(3 + j2)^*}{(4 + j6)^*(3\underline{/70^\circ})}\right]^*$

Step-by-step solution

Step 1 of 4

(a)

Consider the complex fraction, $\dfrac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}}$.

Convert the polar form number into rectangular form.

$$\begin{aligned} 6\angle 30^\circ &= 6\cos 30^\circ + j\sin 30^\circ \\ &= 5.19 + j3 \end{aligned}$$

Convert the exponential form number into rectangular form.

$$\begin{aligned} 2e^{j45^\circ} &= 2\cos 45^\circ + j\sin 45^\circ \\ &= 1.414 + j1.414 \end{aligned}$$

Step 2 of 4

Evaluate the value of $\dfrac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}}$.

$$\begin{aligned} \frac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}} &= \frac{5.19 + j3 + j5 - 3}{-1 + j + 1.414 + j1.414} \\ &= \frac{2.19 + j8}{0.414 + j2.414} \\ &= \frac{(2.19 + j8)(0.414 - j2.414)}{(0.414 + j2.414)(0.414 - j2.414)} \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} \frac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}} &= \frac{0.9066 + j3.312 - j5.2866 + 19.312}{0.414^2 + 2.414^2} \\ &= \frac{20.2186 - j1.97466}{0.1714 + 5.8274} \\ &= 3.37 - j0.3291 \end{aligned}$$

Finally convert the complex number into polar form.

$$\begin{aligned} 3.37 - j0.3291 &= \sqrt{3.37^2 + 0.3291^2} \\ &= \sqrt{11.35 + 0.108}\left\{\angle\left[-\tan^{-1}\left(\frac{0.3291}{3.37}\right)\right]\right\} \\ &= 3.387\angle -5.615^\circ \end{aligned}$$

Thus, the evaluate value of $\dfrac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}}$ is $\boxed{3.387\angle -5.615^\circ}$.

Step 3 of 4

(b)

Consider the complex fraction, $\left[\frac{(15-j7)(3+j2)^*}{(4+j6)^*(3\angle 70^\circ)}\right]^*$

The value of complex conjugate, $(3+j2)^*$, is $3-j2$.

The value of complex conjugate, $(4+j6)^*$, is $4-j6$.

Convert the polar form number into rectangular form.

$$\begin{aligned} 3\angle 70^\circ &= 3\cos 70^\circ + j3\sin 70^\circ \\ &= 1.026 + j2.819 \end{aligned}$$

Step 4 of 4

Evaluate the value of $\left[\frac{(15-j7)(3+j2)^*}{(4+j6)^*(3\angle 70^\circ)}\right]^*$.

$$\begin{aligned} \left[\frac{(15-j7)(3+j2)^*}{(4+j6)^*(3\angle 70^\circ)}\right]^* &= \left[\frac{(15-j7)(3-j2)}{(4-j6)(1.026+j2.819)}\right]^* \\ &= \left(\frac{45-j21-j30-14}{4.104-j6.156+j11.276+16.914}\right)^* \\ &= \left(\frac{31-j51}{21.018+j5.12}\right)^* \\ &= \left(\frac{(31-j51)(21.018-j5.12)}{(21.018+j5.12)(21.018-j5.12)}\right)^* \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} \left[\frac{(15-j7)(3-j2)}{(4-j6)(1.026+j2.819)}\right]^* &= \left(\frac{651.558-j1071.98-j158.72-261.12}{441.7563+26.2144}\right)^* \\ &= \left(\frac{390.438-j1230.7}{467.9707}\right)^* \\ &= (0.8343-j2.6297)^* \\ &= 0.8343+j2.6297 \end{aligned}$$

Finally convert the complex number into polar form.

$$\begin{aligned} 0.8343+j2.6297 &= \sqrt{0.8343^2+2.6297^2}\angle\left[\tan^{-1}\left(\frac{2.6297}{0.8343}\right)\right] \\ &= 2.759\angle 72.39^\circ \end{aligned}$$

Thus, the evaluate value of $\left[\frac{(15-j7)(3+j2)^*}{(4+j6)^*(3\angle 70^\circ)}\right]^*$ is

$$\boxed{\left[\frac{(15-j7)(3-j2)}{(4-j6)(1.026+j2.819)}\right]^* = 2.759\angle 72.39^\circ}.$$